Is the leak-free network running DD?

Causal interventions, not probe scores

YES -- it runs DD's rule, its chaining, and its guard.

```
THE TASK    Colour a 24-node graph, every node of degree 5, with q = 4 colours so that no edge joins
            two like colours. Some nodes are given as hints; the rest must be deduced.

DD          Direct Deduction, the classical 'naked single' solver the model is trained to imitate. Two
            parts at each step. A RULE: eliminate the colours a node's known neighbours already use.
            A GUARD: emit that node only when exactly one colour is left.

THE MODELS  12-layer transformers, each NAMED BY ITS SOLVE RATE -- the fraction of 1500 held-out graphs
            it colours completely and correctly under free generation.
              'leaked'     trained on an encoding whose adjacency ORDER gave the colouring away.
              'leak-free'  trained with that order permuted, so the answer has to be deduced.
```

One-token counterfactuals test the rule and the guard separately:

model	RULE on given hints	RULE on its OWN facts	GUARD withholds	(placebo ceiling)
leak-free 1.0000	1.000	0.980	1.000	0.060
leak-free 0.8540	1.000	0.980	0.824	0.100
leaked 0.9960	0.388	0.008	0.992	0.036
chance	0.250	0.250	--	

Both leak-free models apply the rule and chain it; the leaked one never learned it. The GUARD column does NOT separate leaked from leak-free -- all three withhold -- so passing it is not by itself evidence of DD. What it does separate is the two leak-free twins: the one that solves the task withholds on every case and the other fails 0.176 of them, the same order as its 0.146 solve-rate deficit.

CAVEAT, up front: one task configuration (N = 24, degree 5, q = 4) and ONE CHECKPOINT PER MODEL, no seed replicates -- so every leak-free vs leaked contrast is also two separate training runs.

The task, and the algorithm the model imitates

THE TASK

Colour a random regular graph: N = 24 nodes, every node of degree 5, using q = 4 colours so that no edge joins two nodes of the same colour.

Some nodes are given as HINTS (15 of 24 here); the rest must be deduced.

Each instance is built from a planted colouring by removing hints one at a time, keeping a removal only if DD can still recover everything. So every instance is solvable by DD alone, and instances differ only in how deep the deduction has to go.

DD -- DIRECT DEDUCTION

Sudoku's 'naked single' rule, on a graph: a node becomes determined one round after its known neighbours show all q-1 = 3 colours other than its own. Sweep the nodes, emit every determined one, repeat until none are left.

THE FORCING SET of a node v = the known neighbours whose colours do the eliminating. DD fires on v exactly when that set shows q-1 = 3 distinct colours; v then takes the one colour left over. Every test in this deck is an edit to that set.

REASONING DEPTH T = how many rounds an instance needs, 1 to 6 at this density. The example next needs 2.

Solvable-by-DD is a property of the INSTANCES, not a proof that no other method works -- which is why the deck goes on to rule out the obvious non-deductive routes (reading the leak, page 3; emitting nodes by index order, last page) rather than assuming they are closed.

The leak, and why there are two kinds of model

THE MODEL

A 12-layer decoder-only transformer (d_model 256, 4 heads per layer, MLP 1024), trained from scratch with next-token cross-entropy on DD's own traces: graph, hints, then round by round the nodes DD determines and the colours it gives them.

SOLVE RATE, which also names each model, is scored by FREE GENERATION: the fraction of 1500 held-out instances it colours completely and correctly with no teacher forcing.

THE LEAK

The generator listed each node's neighbours sorted by the neighbour's TRUE COLOUR. So a neighbour's POSITION in the adjacency row encoded its colour, and 61-67% of nodes could be read off with no deduction at all.

Permuting the rows at test time collapses a model that was trained on them: the best such model goes from 0.9975 to 0.1450. The leak was load-bearing, not incidental.

'leaked'	trained on the sorted rows. Scores 0.9960 and needs no reasoning to do it.
'leak-free'	trained with the rows permuted. Two of them, 1.0000 and 0.8540, differing only in training-set size.

THE QUESTION: the leak-free 1.0000 model solves a task whose instances are ALL solvable by DD. Is it running DD, or doing something else that happens to work here?

Every BEHAVIOURAL number from here on is causal: edit one input token, ask whether the output follows DD's answer to the EDITED problem. (The one correlational page, the sparse autoencoder, is labelled as such.)

WHY NOT PROBES: an earlier round of this project read the model's own 'which node is next' features with a trained probe and got 0.545 of nodes right -- while a hand rule that does no deduction at all, 'the unknown node with the fewest unknown neighbours', got 0.632 on the same cases. A decoding score cannot tell those two apart. An intervention can.

What the model actually sees

Every token is one categorical id into a 34-symbol vocabulary (N=24, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-33	`N0 N1 ... N23`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`
`SOLVED`	

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 198 TOKENS (seed 50000000004)

HINTS	`BEGIN_HINTS N2 COLOR_2 N3 COLOR_2 N5 COLOR_2 N8 COLOR_3 N10 COLOR_0 N11 COLOR_1 ... (31 tokens)`
GRAPH	`BEGIN_GRAPH N0 N16 N5 N18 N13 N14 N1 N8 N13 N2 N11 N19 ... (145 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 N9 COLOR_1 N12 ... (22 tokens)`

Loss is on the REASONING section only: 2 rounds, 9 deductions, 18 target tokens. Everything before BEGIN_REASONING is the problem statement.

DD is sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid.

One real reasoning trace

15 hints of 24 nodes, then 2 rounds of simultaneous deduction. Hints: N2=COLOR_2 N3=COLOR_2 N5=COLOR_2 N8=COLOR_3 N10=COLOR_0 N11=COLOR_1 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND            N0 COLOR_1 N1 COLOR_0 N4 COLOR_0 N6 COLOR_1 ...
  8 deduced
BEGIN_ROUND            N7 COLOR_2
  1 deduced
SOLVED
```

WHY ONE DEDUCTION FIRES

node N0, neighbours N16 N5 N18 N13 N14

their colours known before this round:

```
   [0, 2, 3]  (3 = q-1 of the 4)
```

only colour left: 1

```
so the trace emits  N0 COLOR_1
```

The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

The intervention edits ONE colour token: in this section (the CHAINED EDIT) or back in the hints (the HINT EDIT).

T=2 rounds, the median of 40 draws from this config (range 1-6). The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

The intervention: one token, two different correct answers

```
DD forces node v because v's FORCING SET -- its known neighbours -- already uses q-1 = 3
of the 4 colours. Exactly one of them, the neighbour w, holds colour a. So v must take
the one colour left over: c. Recolour w to c, and now a is the colour nobody holds, so
DD's answer for v changes from c to a.

   clean     ... BEGIN_ROUND  v  ->  [c]        w holds a, so c is v's only option
   corrupt   ... BEGIN_ROUND  v  ->  [a]        w recoloured to c, so a is now free

The ONLY difference between the two prompts is w's single colour token, and the code
checks that mechanically on every case rather than assuming it. Scored as 'moved to
DD's new answer' -- did the prediction at v become a? Not 'is it still accurate': the
correct answer CHANGES, so a memorised output scores zero. The model must pick one of
q = 4 colours, so chance on this test is 0.25.

Two versions, differing only in WHERE the edited token sits:

   HINT EDIT     w is a HINT, handed to the model in the prompt, so the edit lands
                 before any reasoning. Tests the elimination rule on GIVEN facts.

   CHAINED EDIT  w was DEDUCED by the model itself in an earlier round, so the edited
                 token sits in the REASONING section. Tests elimination against a fact
                 the model produced -- propagation, not one rule application.
```

One weak direction, stated up front: editing an emitted deduction makes the prefix inconsistent, so a model that DETECTS the edit and keeps its own answer scores as 'does not chain'. Those cases are counted separately, as 'kept its old answer'.

The converse intervention: break the forcing, and v should vanish

Both tests on the previous page SELECT a node that DD forces and then REDIRECT the forcing. A model that simply wrote a colour for whichever unknown node came next in index order would pass them both. So break the forcing instead of redirecting it -- the WITHHOLD test:

   recolour w to a colour that ANOTHER known neighbour of v already holds.

Now the freed colour a is held by nobody, v's forcing set covers only q-2 = 2 colours, v has TWO candidates, and DD does not emit v in this round at all.

The measurement moves ONE TOKEN TO THE LEFT: from the logits that predict v's COLOUR to the logits that predict WHICH NODE comes next. Does the model stop naming v?

```
clean      ... BEGIN_ROUND  [v]  ->  c       DD emits v, and so should the model
broken     ... BEGIN_ROUND  [ ? ]            v is no longer determined
```

PLACEBO, run on every single case: the same kind of one-token edit, but on a hint that is NOT a neighbour of v. v stays forced and DD's round is unchanged, so the model should still name v. That is the ceiling the broken rate is read against -- without it, a low rate would only show that edits perturb the node stream.

Two caveats, both stated before the result. (1) If the placebo collapsed too, the test would say nothing. It does not collapse, but it does not reach 1.000 either (0.90-0.96 across models), so a few points of every rate are edit perturbation -- which is why all three models are read against it. (2) ALL THREE MODELS PASS THIS TEST, the leaked one included. Detecting that a node stopped being determined is evidently easier than applying the rule, so the withhold test is not evidence of DD on its own. Its job is to separate the two leak-free twins.

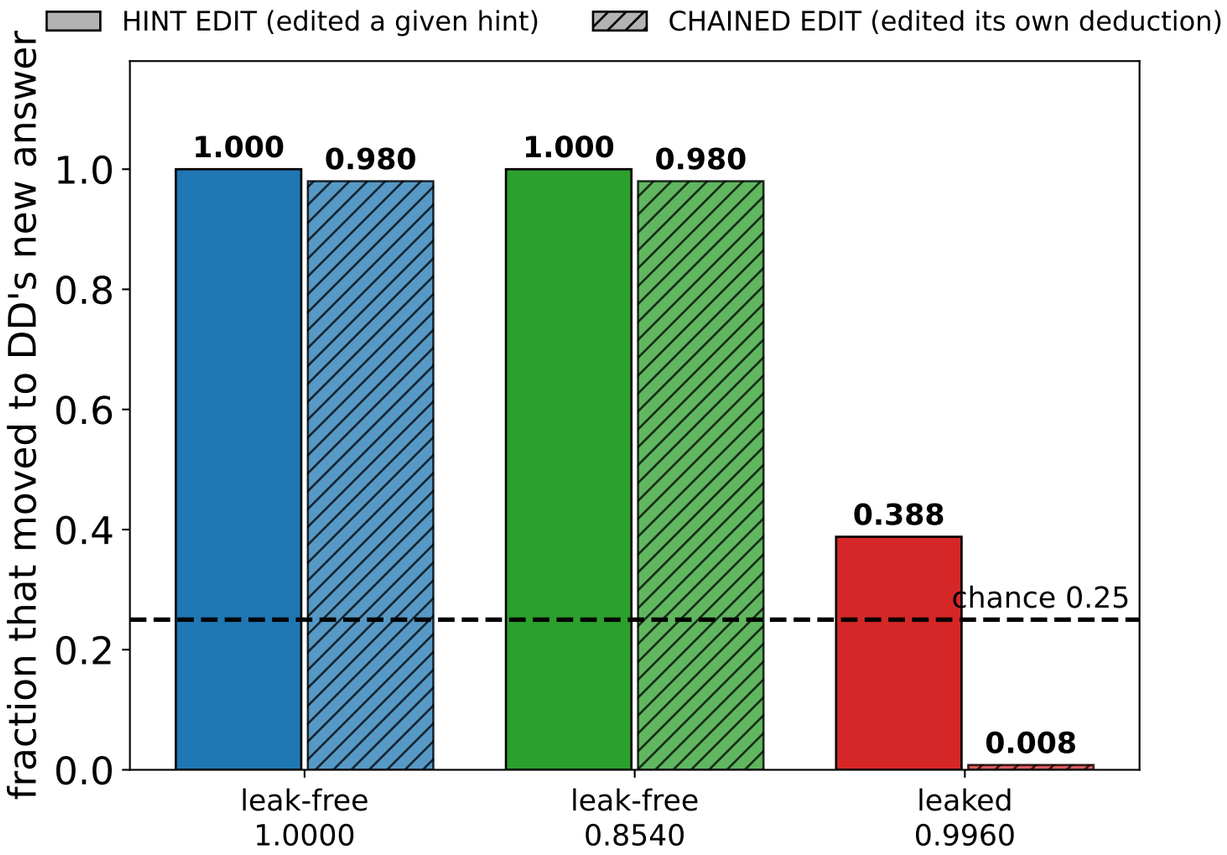
Every causal measurement

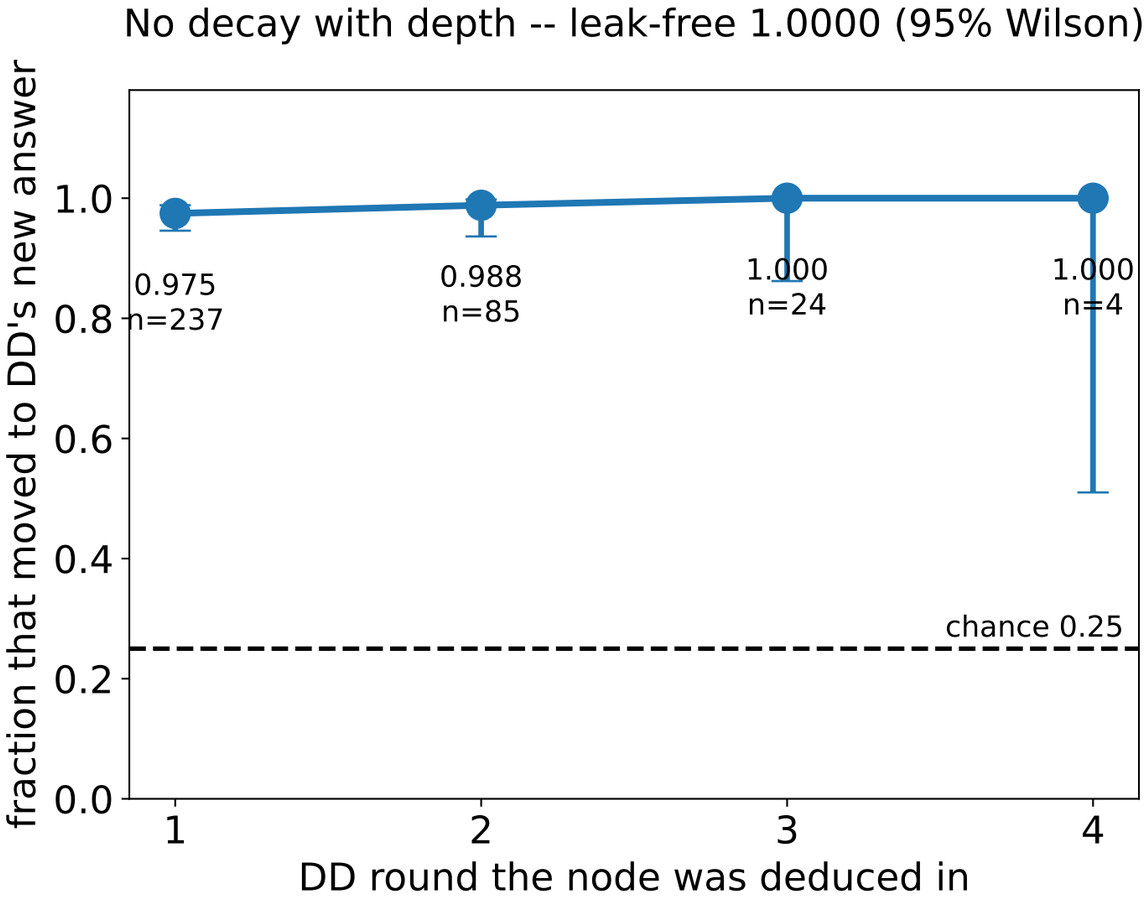
model	test	n	clean	moved to DD's new answer	kept its old answer	logit gap (rule) or placebo moved (guard	verdict
leak-free 1.0000	hint edit	350	1.000	**1.000**	--	-15.5 -> +15.7	uses the rule
leak-free 1.0000	chained edit	350	1.000	**0.980**	0.020	--	chains
leak-free 1.0000	withhold	350	1.000	**1.000**	0.000	placebo 0.060	guards
leak-free 0.8540	hint edit	250	1.000	**1.000**	--	-15.0 -> +14.9	uses the rule
leak-free 0.8540	chained edit	250	1.000	**0.980**	0.008	--	chains
leak-free 0.8540	withhold	250	0.936	**0.824**	0.176	placebo 0.100	guards
leaked 0.9960	hint edit	250	1.000	**0.388**	--	-14.5 -> -2.0	near chance
leaked 0.9960	chained edit	250	1.000	**0.008**	0.992	--	keeps its own answer
leaked 0.9960	withhold	250	1.000	**0.992**	0.008	placebo 0.036	guards

Every column is scored so that HIGH = followed DD, placebo included. A low 'moved' next to a high 'clean' means the model is right for some reason other than the elimination -- exactly the leaked model's profile on its two rule rows. Note its WITHHOLD row: it guards as well as anyone, so that test alone does not distinguish a model that runs DD from one that does not.

LOGIT GAP = logit(DD's answer to the EDITED problem) - logit(DD's answer to the CLEAN one) at the position that predicts v's colour, in raw logits (log-odds), averaged over cases; the arrow spans clean prompt -> edited prompt, so a sign change is a reversal of preference. On the withhold rows 'moved' = it stopped naming the node, 'kept its old answer' = it named it anyway, 'placebo moved' = how often the control edit also stopped it: the noise floor. 'clean' differs between a model's own rows because the conditions do -- the rule rows score the colour emitted for v, the withhold rows whether v is named at all. n is 350 or 250 by when each run was submitted, not by filtering; paired tests use the 250 shared cases.

The rule holds on given facts, and on facts it derived itself

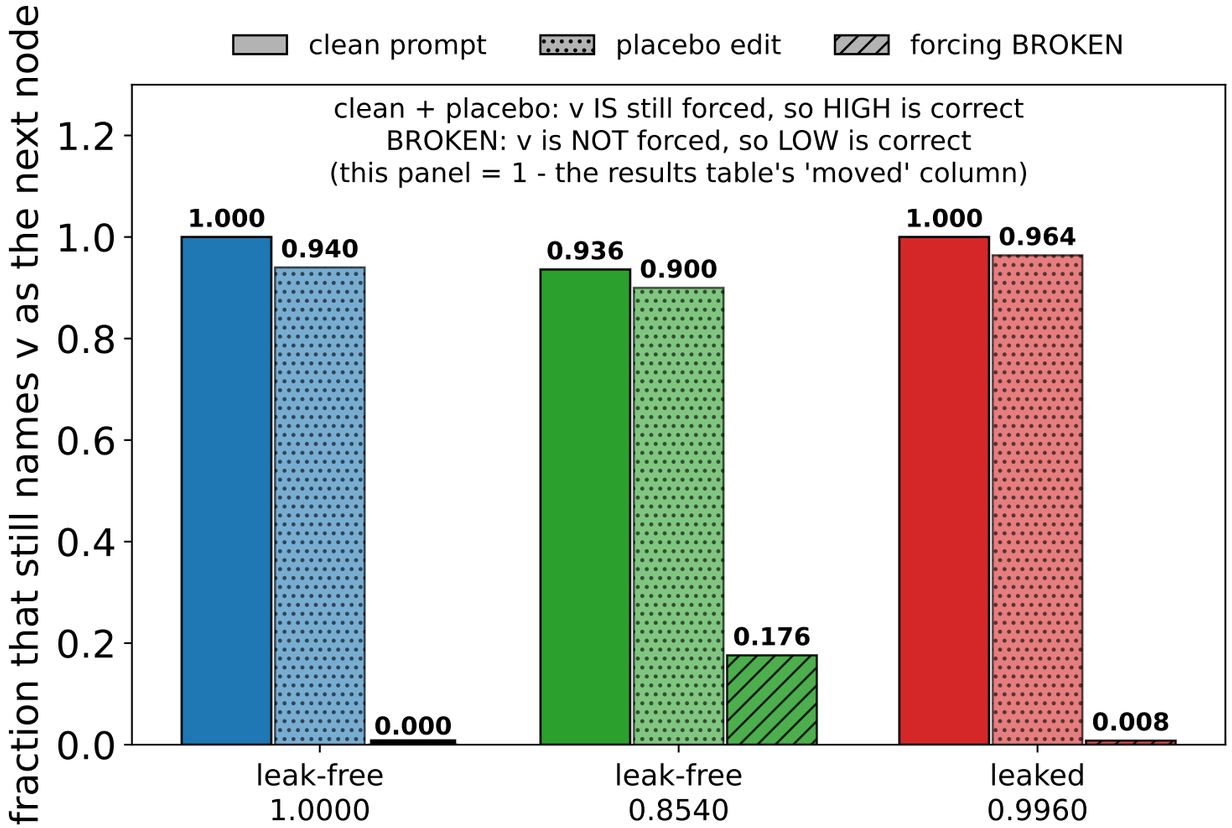


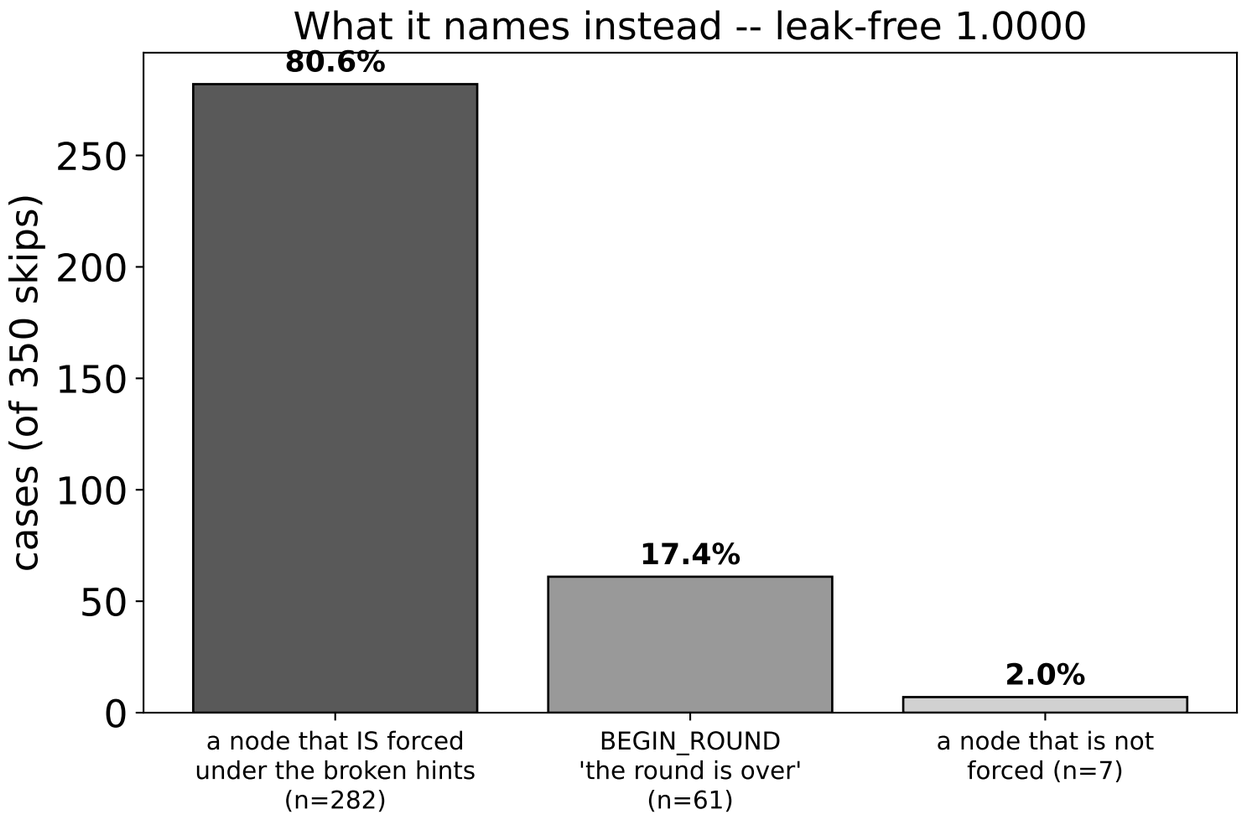


If the model chained one step but not three, round 1 would be high and the later rounds would fall off. It does not happen.

Right panel splits the chained test by how deep the edited deduction sat. Deep bins are small because deep chained forcings are rare in these instances, not because anything was filtered -- hence the intervals, wide at rounds 3-4 (n=24 and n=4) and still clear of chance. Rounds 5-6 are absent because no case in the sample had its binding neighbour deduced that late.

And it withholds: break the forcing and it stops naming the node

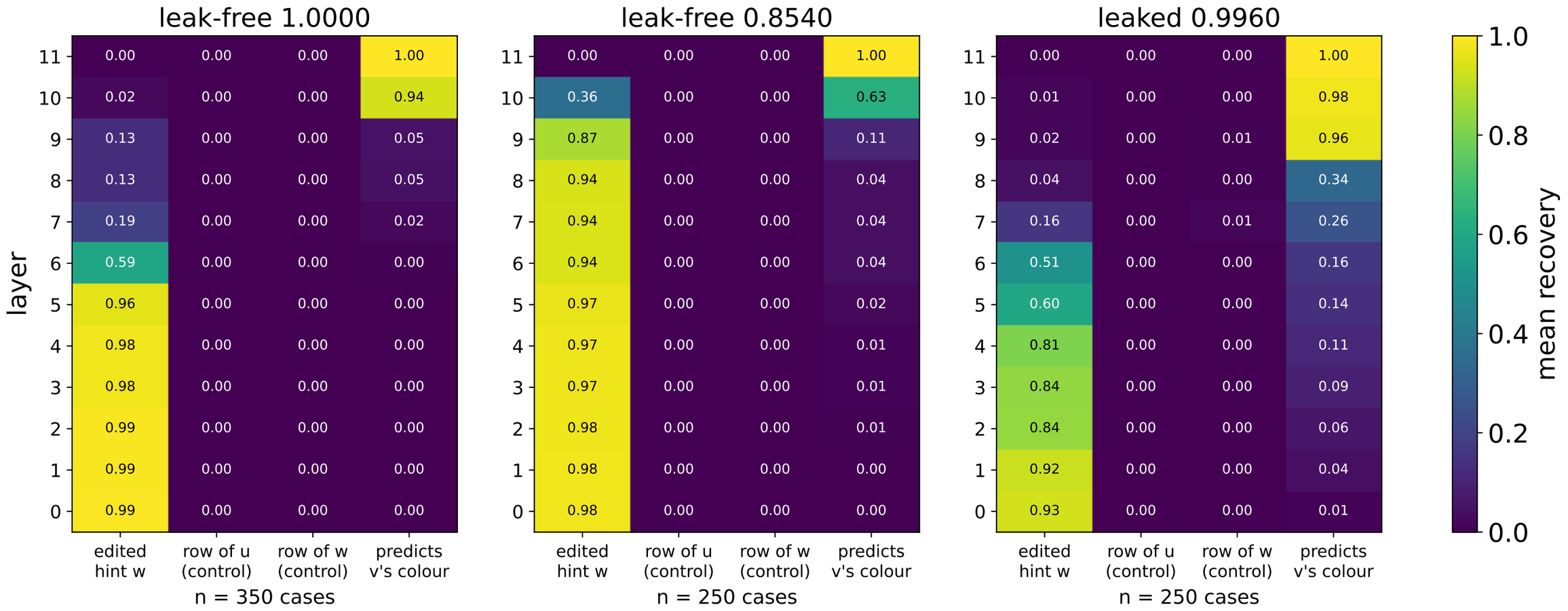




The two leak-free twins differ here and nowhere else: McNemar paired, 44 cases where the 0.8540 twin named v and the 1.0000 model did not, 0 the other way, p = 1.1e-13. It does not just go quiet either: of the 350 cases it withholds, it names another node that IS forced under the broken hints, or says BEGIN_ROUND -- DD's own answer when nothing else in the round is determined.

Same cases as the hint-edit test, and the case lists are byte-identical across models, so the arms are paired instance for instance. Conditioning on cases the twin gets right on the clean prompt gives 0.188 (n=234). Its 0.176 guard gap is the same order as its 0.146 solve-rate gap -- different measurements on different samples, so a consistency check, not a decomposition. The leaked model withholds too (0.992): this test separates the twins, not leaked from leak-free.

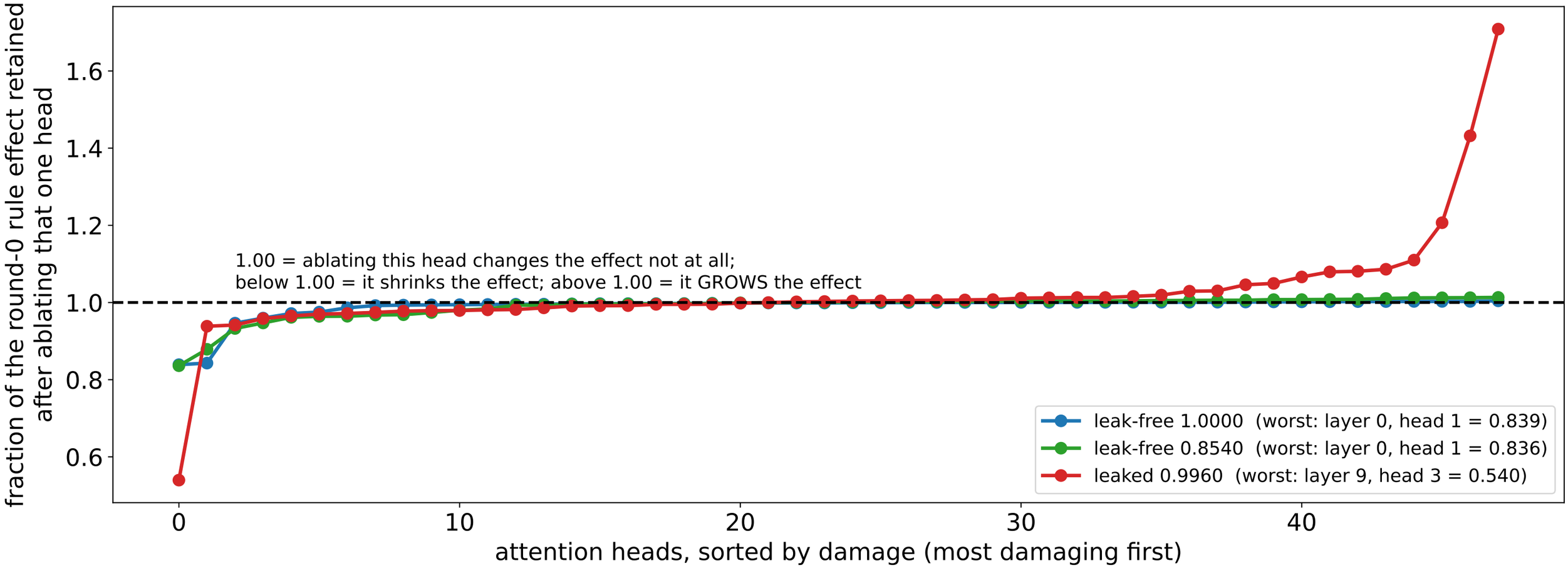
Where the recoloured fact travels (activation patching)



RECOVERY = copy one activation from the clean run into the corrupted run at that (layer, position) and measure how much of the flip it undoes: 1.00 = that one activation carries the whole edited fact, 0.00 = none of it. w is the neighbour the edit recolours, u is another known neighbour of v; both graph rows are controls, and 'predicts v's colour' is where the answer is read off.

Read upwards: the edited colour is held AT the edited hint through L5, handed off between L6 and L10, and has arrived at the prediction position by L10. The graph rows read 0.00 at every layer in every model -- the deduction never routes through the adjacency table. L11 at the prediction position is the built-in positive control and must read 1.00; it does. The 1.0000 model RELEASES the fact from the hint token about 3 layers earlier than its 0.8540 twin (crossing half recovery at L6.2 against L9.7), though both have it at the prediction position by L10, so the difference is in how long it is held, not in where it lands. In the leaked panel recovery is a fraction of an effect that is near chance to begin with (its flip rate is 0.388), so its numbers localise a sub-threshold sensitivity, not a rule it applies.

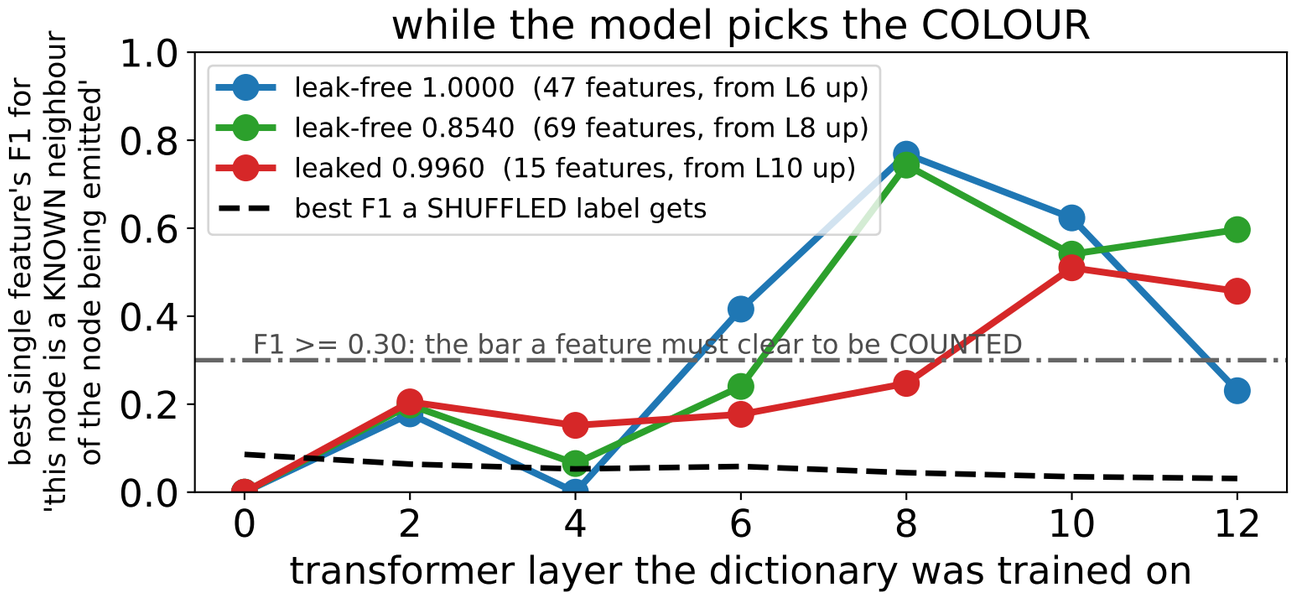
No single head carries the rule: removing the worst one leaves 84% of it

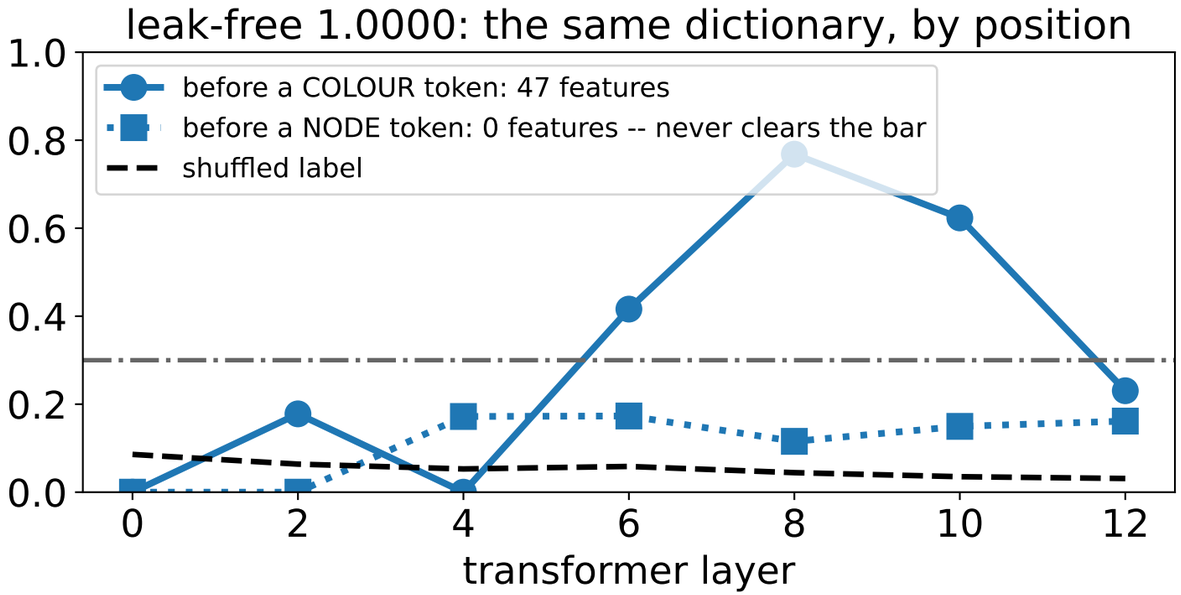


The leaked model's curve only looks steeper because its effect is near chance to begin with: a fraction of almost nothing.

Each point is one of the 48 heads (12 layers x 4 heads), zeroed one at a time, averaged over the same 350 (leak-free 1.0000) or 250 (the other two) hint-edit cases as the rule test. Values above 1.00 mean ablating that head makes the model follow the edit MORE, which happens when a head is pulling toward the pre-edit answer. So there is no single circuit to point at: the rule is distributed across heads, and this is a negative result about localisation, not about DD.

It carries sparse features for DD's own forcing set, peaking at L8 -- inside the L6-L10 window where the causal patch hands the fact off



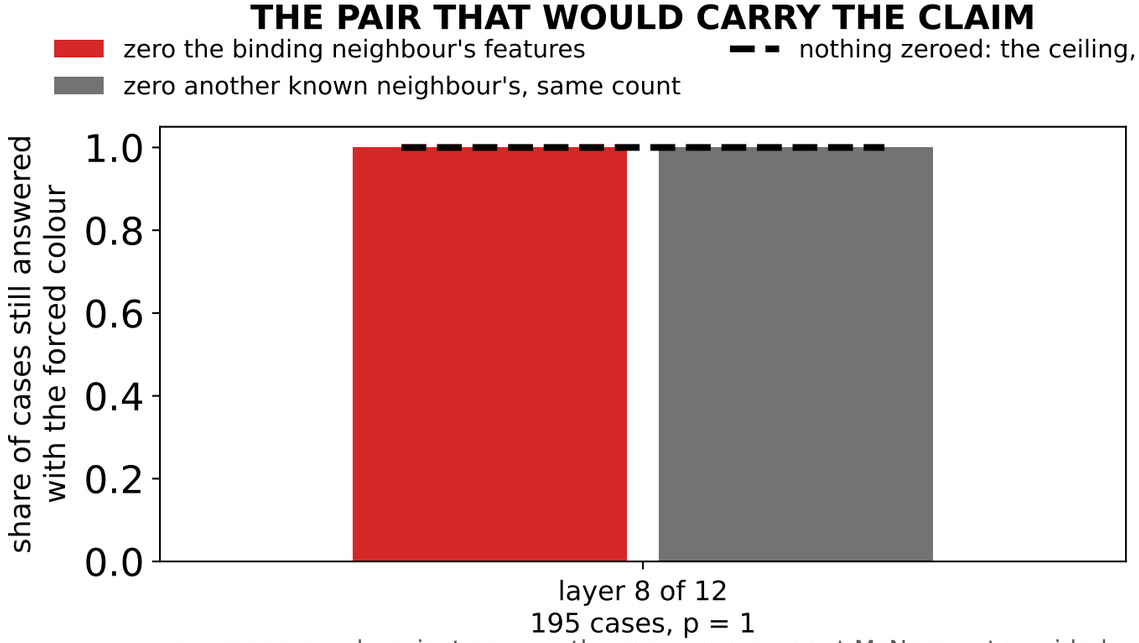


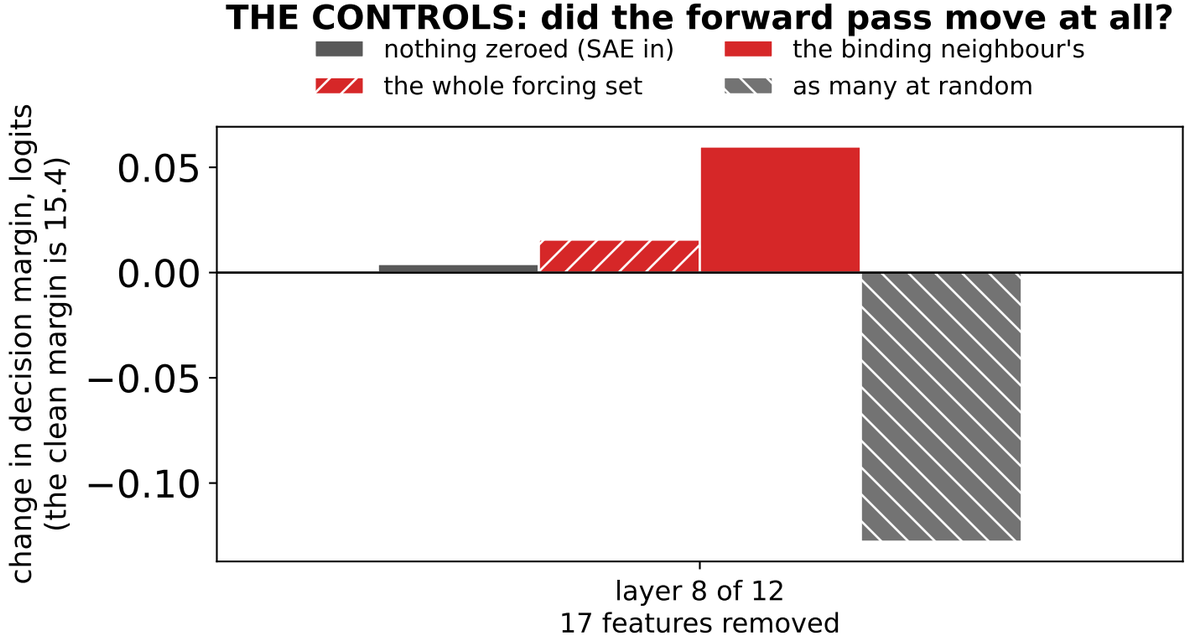
A feature is COUNTED only when it matches one of DD's own predicates at F1 >= 0.30 (F1 = the harmonic mean of precision and recall) AND beats a shuffled-label null. Three things to read: (1) the forcing set is encoded where DD's rule needs it -- at the token that predicts the COLOUR, and never at the token that picks the node; (2) the 1.0000 model is NOT more interpretable than its 0.854 twin, which is what their equal rule scores already said; (3) THE LEAKED MODEL ENCODES THE FORCING SET TOO -- 15 features, but only at L10-L12, after the window where the leak-free model uses it. Encoding it is not using it: that is why this page is not the evidence, and why pages 6-11 are.

CORRELATIONAL -- this page says WHERE the structure sits, not whether the model uses it. Sparse autoencoder, 4096 features per layer, ~64 active per token, trained on 53,200 activations per layer, then scored separately on the 26,600 positions before a colour token and the 26,600 before a node token, so counts compare within a mask only. Layer 12 is the residual stream after the last block, which is why this axis runs one further than the patching page's. One dictionary per layer per arm, one seed.

Nothing we remove moves the answer -- not the binding neighbour's features, not the whole forcing set, not a random set of the same size. This test has no sensitivity to report from.

The binding neighbour is the one whose colour rules out the node's last alternative; its fellow known neighbours are just as known, but not decisive.





p compares red against grey on the same cases, exact McNemar, two-sided.
A case counts only if it owns features for both kinds of neighbour.

THE DESIGN: every bar is read against the reconstruction condition (the SAE's own output spliced back in, nothing zeroed), so reconstruction error is netted out rather than charged to the ablation. The two bars on the left remove the same number of features, of the same kind, at the same position, in the same layer, on the same cases; they differ only in which neighbour the features refer to, the one whose colour does the forcing or another known neighbour whose colour does not. Only the model reading that neighbour's colour to eliminate a candidate can open a gap between them. The powered replication -- the same test widened to every feature that beats its null, 195 paired cases at layer 8 -- gives +0.00, p = 1. Prediction (d) is falsified: at the token that picks the node the widened gate did find features to ablate (35 at layer 8) where finding 21 found none, and removing those changed nothing there either.

READ IT AS: inconclusive -- and the control says so before the headline does. Removing the whole forcing set had to cost more than removing a count-matched random set; at layer 8 both cost nothing, 1.000 against a 1.000 baseline, and the paired comparison has 195 cases with not one discordant answer, 95% CI [-0.015, +0.015]. The clean decision margin is 15.4 logits; the largest of these ablations shifts it by 0.13, under 0.8% of the margin, and removing the forcing set shifts it the wrong way, towards MORE confidence (+0.02). So this is not evidence that the features go unused. It is a test in which zeroing 17 of 4096 features at one token cannot perturb a decision this overdetermined -- no sensitivity, so no null to report. The verdict on the next page rests on the interventions, which do move the answer.

The first run hooked the wrong block (dictionary L reads encoder block L-1) and its numbers are discarded, not shown here. Dictionaries: the model that solves the task (solve_rate 1.0000), 4096 features a layer, one seed. The 300 cases are held-out first-round deductions whose forced colour one neighbour's recolouring can flip; features come from the previous page's threshold.

The verdict, and what it is not

Working picture: sweep nodes in index order; emit one only when its known neighbours leave it a single colour, counting the ones it just derived. That is DD's propagation step.

WHAT IT DOES DO (causal unless marked)

* Applies DD's elimination rule: 1.000 on 350 one-token counterfactuals.

* Chains it off its OWN deductions: 0.980, no decay from round 1 to round 4.

* Guards it: break the forcing and it stops naming the node 1.000 of the time, while a placebo edit stops it only 0.060. But ALL THREE models pass this one, so passing it is not by itself evidence of DD.

* Its 0.854 twin guards only 0.824, a gap the same order as that twin's 0.146 solve_rate deficit -- and the first test here to separate the two TWINS at all.

* Localised in depth: edited hint -> L5, handoff L6-L10, prediction by L10.

* CORRELATIONAL: sparse features for DD's forcing set peak at that same L8 -- but so do the leaked twin's. Ablating them moves nothing; nor do the controls (previous page).

WHAT IT DOES NOT DO

* Choose the node ORDER by deduction. It emits nodes in ascending index order in 100% of rounds. That is also DD's own convention, so it proves nothing either way -- but the guard test does show that WHICH nodes it emits tracks determinacy.

* Route through the adjacency table: the graph rows recover 0.000 at every layer.

* Concentrate the rule anywhere we can point at: ablating the worst of the 48 heads one at a time still leaves 84% of the effect. The rule is distributed, so this deck can say THAT it runs DD, not which heads do.

* The leaked twin never learned the rule (0.388, chance 0.250). Shown an edited deduction it keeps its own answer 99.2% of the time: it reads the adjacency-order leak, not the prefix.

LIMITATIONS: one checkpoint per model and no seed replicates, so every leak-free vs leaked contrast is also two separate training runs; one task configuration (N = 24, degree 5, q = 4), so nothing here is a claim about depth or scale; the per-depth and per-degree breakdowns have bins as thin as a few dozen cases, where the error bars overlap everything; the guard gap and the solve_rate gap are different measurements on different samples, so their agreement is a consistency check, not a decomposition; and the sparse-autoencoder pages are correlational: ablating those features moved no answer at all, controls included, so that test is inconclusive.